

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering
B.E. Artificial Intelligence and Machine Learning

Bengaluru, India

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Designed and built a resilient **Node.js/Puppeteer workflow engine** for MCA company-registration filings, turning browser-driven steps into repeatable backend operations.
- Integrated **public-site API lookups**, retries, session recovery, and operator checkpoints so partially completed workflows could resume instead of restarting, reducing manual filing time by **85%**.
- Implemented an **AI-assisted drafting flow** for company-name suggestions, object descriptions, and filing fields, with inspectable outputs and deterministic validation before execution.
- Containerized backend services with **Docker**, deployed them on **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Worked across InVesalious, an open-source **3D medical-imaging and neuronavigation application**, integrating FastSurfer CNN inference for **95 anatomical brain regions** with PyTorch/TorchScript and ONNX/TinyGrad paths.
- Handled **DICOM and volumetric MRI workflows** alongside NIfTI/MGZ data, including voxel conformation, orientation-sensitive preprocessing, plane-specific models, label remapping, LUT mapping, generated masks, and 3D surface inspection.
- Debugged **model-output orientation, label correctness, GPU/CPU paths, progress reporting, and 3D inspection** so segmentation results remained consistent across inference paths.
- Studied model architecture and runtime behavior while converting/integrating inference paths, then reduced processing time by **87%** through async/parallel execution with GPU/CPU, output-correctness, and QC checks.

Open Source

Contributions 

- Contributed maintainer-reviewed patches to **Invesalious** around medical-imaging artifact correctness and parser safety, with additional work in scientific data codebases.
- Also contributed to **Kuadrant MCP Gateway, Kyverno, Tiled, CRIU, IOOS, and VideoLAN**, covering backend failover behavior, security validation, data and parser correctness, and compatibility fixes.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, Next.js, Docker*

GitHub 

- Built an **AI Reliability & Evaluation Platform** for **traceable validation runs**, replayable runs, release gates, promotions, audit logs, and project-scoped API keys.
- Designed **reproducible worker execution** with database leases, content-addressed artifacts, signed manifests, OTLP/OpenInference ingestion, and **Postgres-backed metadata**.

Postificus | *Go, Echo, React, Vite, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

GitHub 

- Built a **full-stack workflow app** with **REST APIs**, auth/settings endpoints, **PostgreSQL persistence**, Redis cache/autosave, RabbitMQ jobs, S3-compatible storage, and live publication status.
- Added **retries, DLQs, circuit breakers**, health checks, and **Prometheus metrics** so long-running API/browser workflows stayed observable and recoverable under unreliable integrations.

Technical Skills

Languages: Python, TypeScript, Go, JavaScript

Frameworks: PyTorch, FastAPI, Next.js, React, SQLAlchemy, VTK, wxPython

Tools: ONNX, NumPy, PostgreSQL, Redis, Docker, OpenTelemetry, GitHub Actions